Can AI Competitors Cooperate on Safety?

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- Stopping first sacrifices competitive position while shared danger continues until the rival also stops.
- Faster monitoring can rule out perpetual racing, provided firms respond optimally when they learn that a rival has stopped.
- An AI can serve its own developer perfectly and still contribute to a dangerous race.

Who Wants to Be the First to Slow Down?

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The question

Can a dangerous race persist because a rival would not promptly recognize and reciprocate a decision to stop?

Even an AI that perfectly optimizes its own firm's objective faces this coordination problem.

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$$x_i(t) = \min\{t, \tau_i\},$$

where $\tau_i \in [0, \infty]$ is firm i's stopping time; $\tau_i = \infty$ means it never voluntarily stops.

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- Stopping freezes capability, but the firm keeps earning profits from what it has already developed.
- Stopping is irreversible in this model.
- Disaster ends both firms' profits. The risk disappears only after *both* firms stop developing.

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If both develop at the same speed, they remain tied and each earns profit at rate one.
If the firms are tied and firm i stops while its rival develops for another s units of time, their profit rates become

$$\pi_i = e^{-s}, \qquad \pi_j = e^{s}.$$

The first stopper falls behind. Stopping together preserves equality.

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We take $\rho = 1$ to keep the calculations simple. Firm i's payoff is its expected discounted total profit:

$$\mathbb{E}\left[\int_0^T e^{-t}\pi_i(t)\,dt\right].$$

Here T is the catastrophe time, or ∞ if none occurs.

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Expected discounted profits during development have three factors:

$$\underbrace{e^{-s}}_{\text{discount}} \quad \underbrace{e^{-2s}}_{\text{survival}} \quad \underbrace{\pi_i(s)}_{\text{profit rate}} .$$

Discounting reflects impatience; survival reflects disaster risk. The chosen rates $\rho = 1$ and $\lambda = 2$ are not empirical estimates.

Both Firms Prefer Stopping Together

If both stop now, each keeps profit rate one forever, without disaster risk:

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Both prefer joint stopping. But would either firm benefit from stopping *first*, while the rival continues?

A Rival's Stop Is Observed Only After a Delay

If a firm stops, its rival receives a conclusive report after an independent delay D, with

$$\mathbb{P}(D > s) = e^{-\omega s}, \qquad s \geq 0.$$

This is an exponential delay with rate $\omega > 0$. Its mean is $1/\omega$: larger ω means faster detection.

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We exclude continuous, exact profit observations: knowing its own capability and profit rate would let a firm recover its rival's capability,

$$x_j(t) = x_i(t) - \log \pi_i(t),$$

and detect a stop without waiting for the report.

Keep Developing Until a Stop Report Arrives

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The equilibrium test

Holding the rival's strategy fixed, can a firm improve its expected payoff by changing its own stopping rule?

Slow Monitoring Can Sustain a Dangerous Race

Theorem (In this two-firm model)

The proposed strategies form a Nash equilibrium if and only if

$$0 < \omega \leq \frac{1}{2}.$$

In this equilibrium, neither firm voluntarily stops, each has expected payoff $1/3$*, and catastrophe occurs with probability one. Both firms would prefer immediate joint stopping, which gives each payoff one.*

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We will verify two claims:

1. After learning that the rival has stopped, stopping too is optimal.
2. Before any report arrives, no alternative stopping plan improves on continuing to develop until a stop report arrives, exactly when $\omega \leq 1/2$.

Waiting Raises Profits but Prolongs Disaster Risk

Suppose we receive a report that the rival has stopped. Let $b > 0$ denote our profit per unit of time when the report arrives—the quantity previously written as $\pi_i(t)$. Measure elapsed time s from this moment.

Compare stopping now with developing until time h, then stopping. The choice $h = \infty$ means never stopping.

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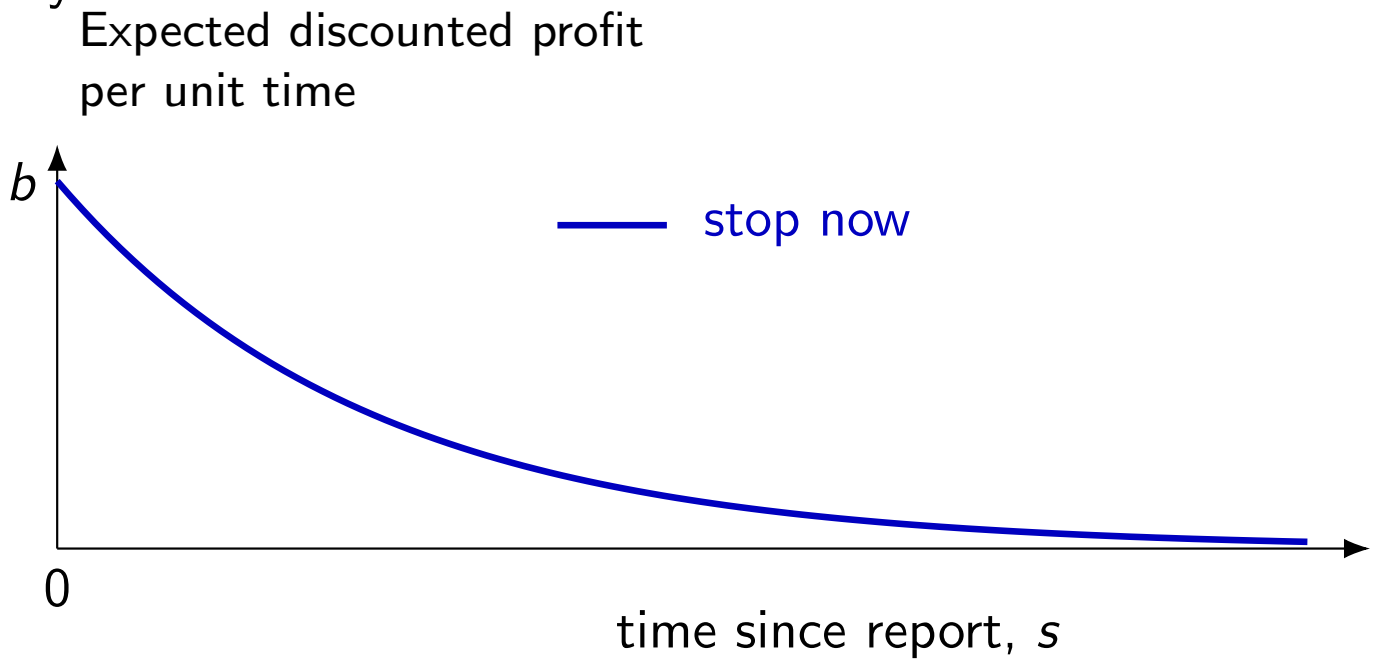
By date s, the waiting rule has kept us developing for $\min\{s, h\}$ units of time:

At date s	Stop now	Stop at h
Profit rate if alive	b	$be^{\min\{s,h\}}$
Probability of being alive	1	$e^{-2\min\{s,h\}}$

Multiply the two entries in each column by the common discount factor e^{-s} to obtain that date's expected discounted profit per unit time.

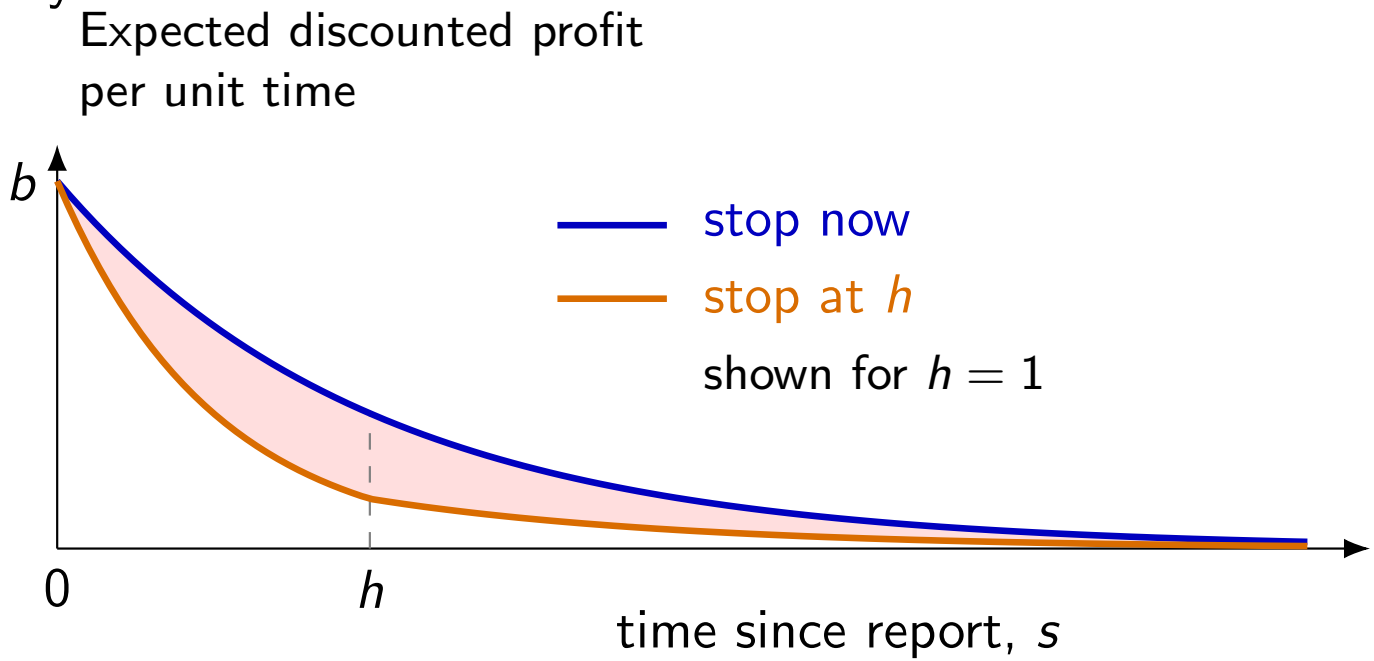
Waiting Reduces Expected Profit at Every Date

Total payoff is the area under the curve over all future dates.



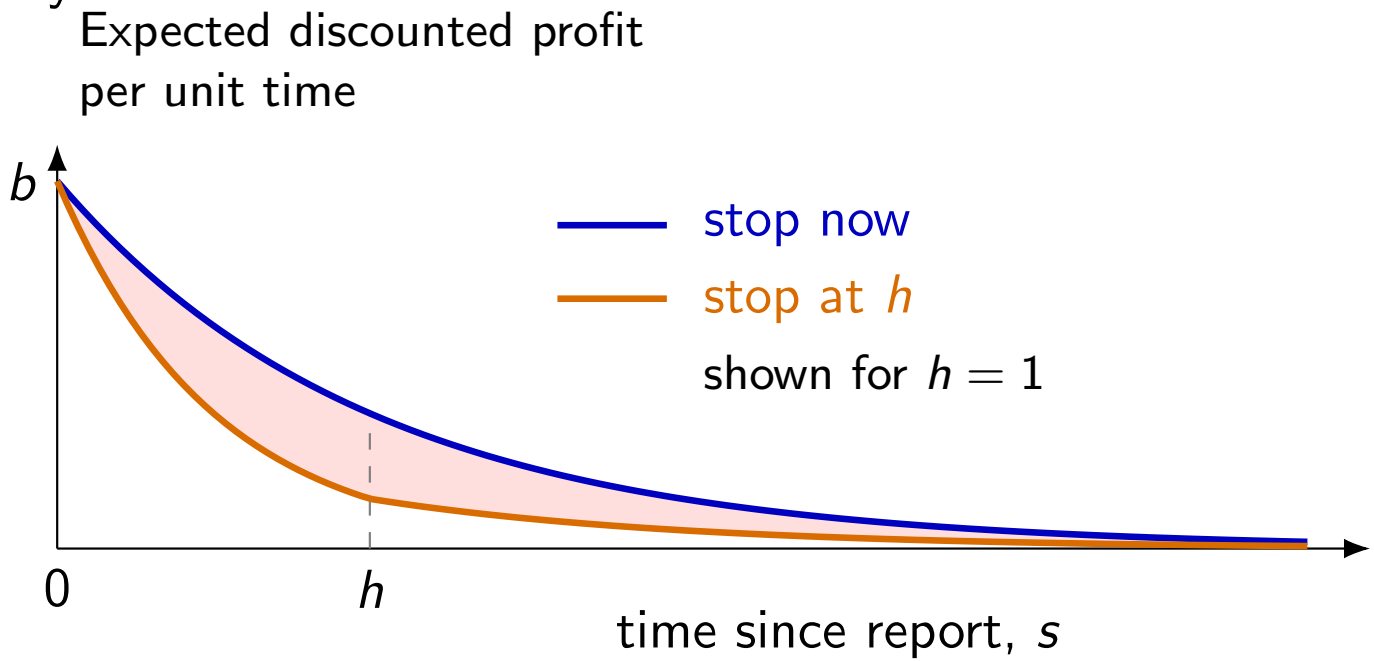
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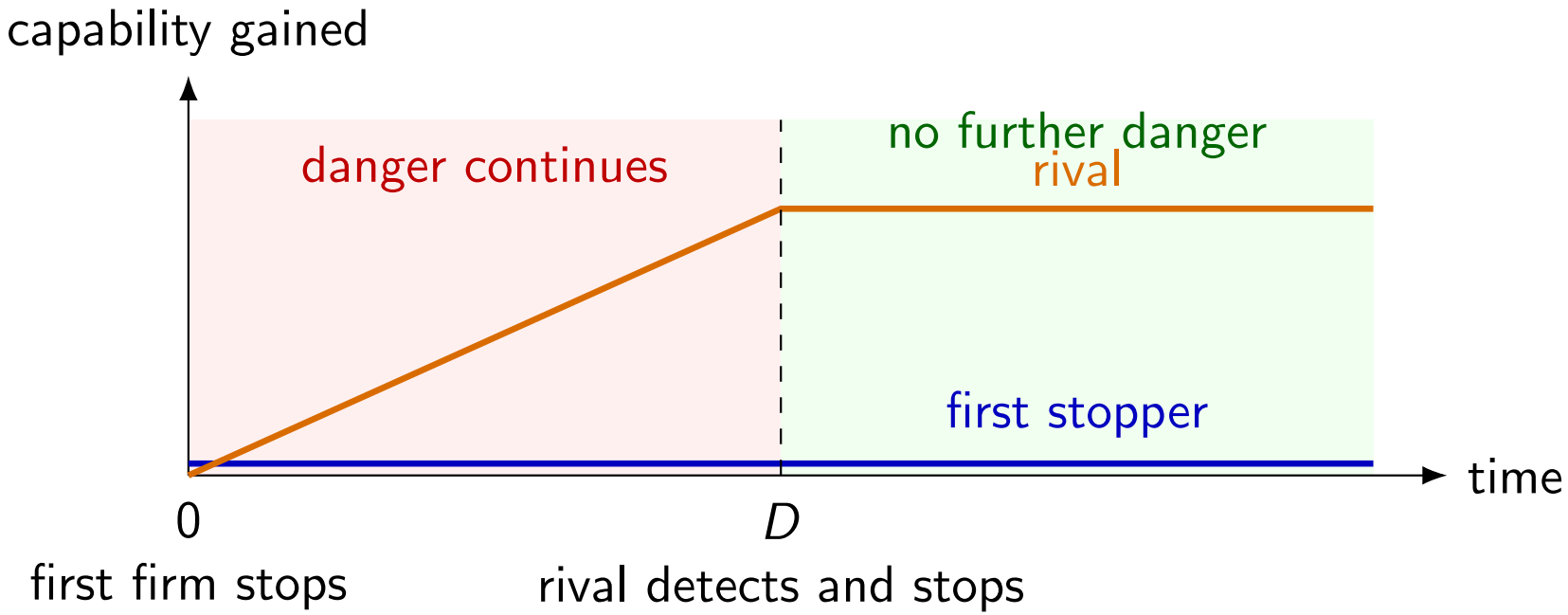
For every $s \geq 0$, $b > 0$, and waiting time h,

$$\underbrace{be^{-s}e^{-\min\{s,h\}}}_{\text{stop at } h} \leq \underbrace{be^{-s}}_{\text{stop now}} .$$

So stopping now maximizes total payoff. Randomizing the wait cannot help; the firm need not know b, because the comparison holds for every b.

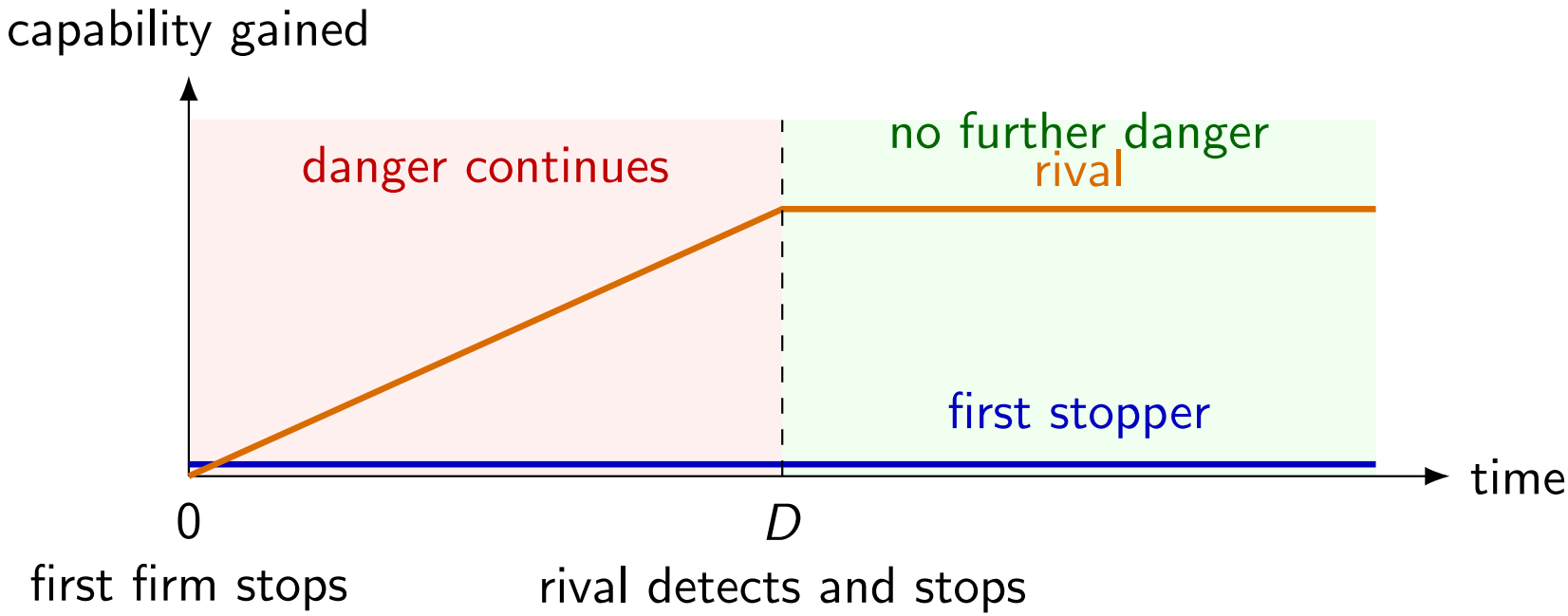
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Before detection, the first stopper loses ground *and* remains exposed to catastrophe. After detection, its competitive disadvantage remains, but disaster risk ends.

The Payoff From Stopping First Has Two Parts

Let V_{first} be our expected payoff if we stop now, while tied, and the rival stops upon receiving the report.

Before the report

The rival keeps developing. Our profit rate falls, and catastrophe can still occur.

After the report, if alive

Both firms have stopped. Our profit rate stays lower, but disaster risk ends.

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The expected discounted profits from these two periods add up to

$$V_{\text{first}} = \underbrace{\frac{1}{4+\omega}}_{\text{before the report}} + \underbrace{\frac{\omega}{4+\omega}}_{\text{after the report}} = \frac{1+\omega}{4+\omega}.$$

The full calculation is in the backup slides.

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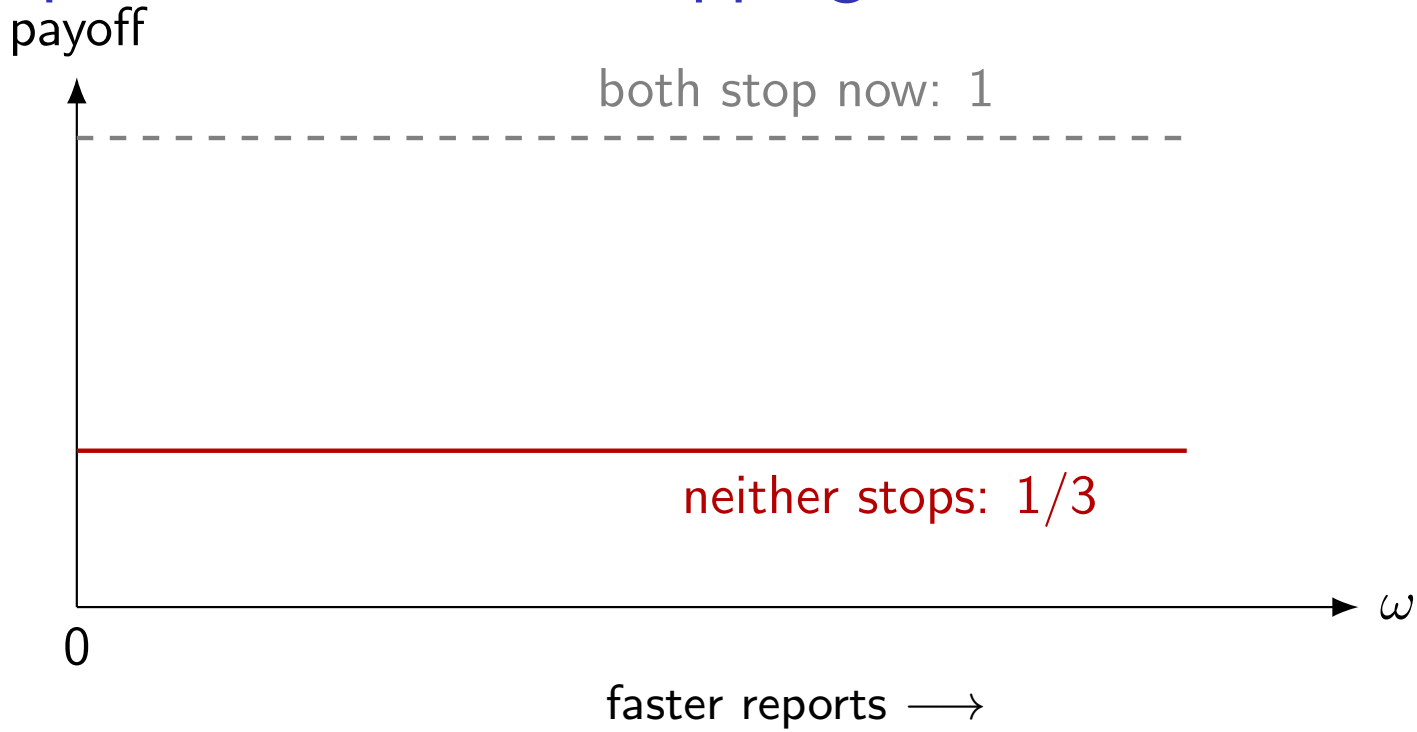
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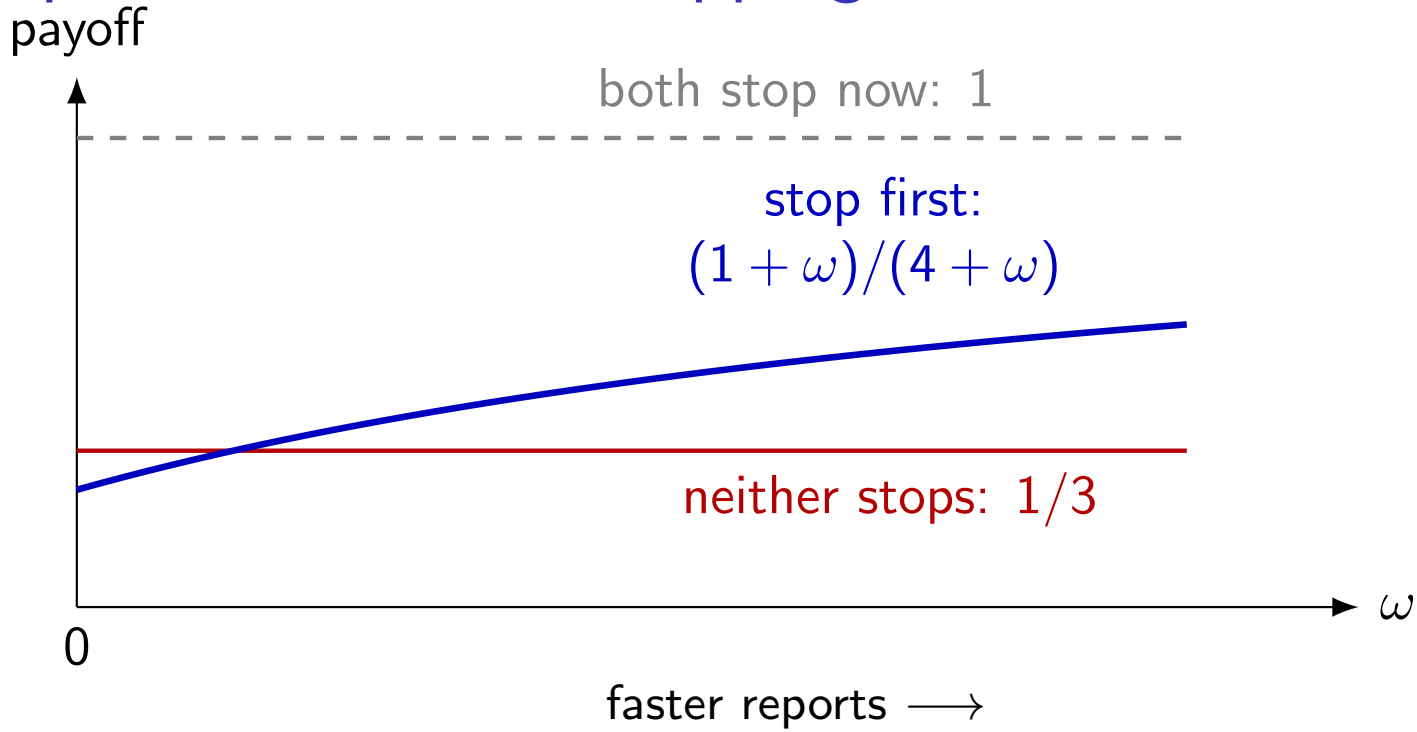
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The comparison is with $V_{\text{race}} = 1/3$, the payoff if neither firm stops. Does stopping first give a higher expected payoff?

Faster Reports Can Make Stopping First Better

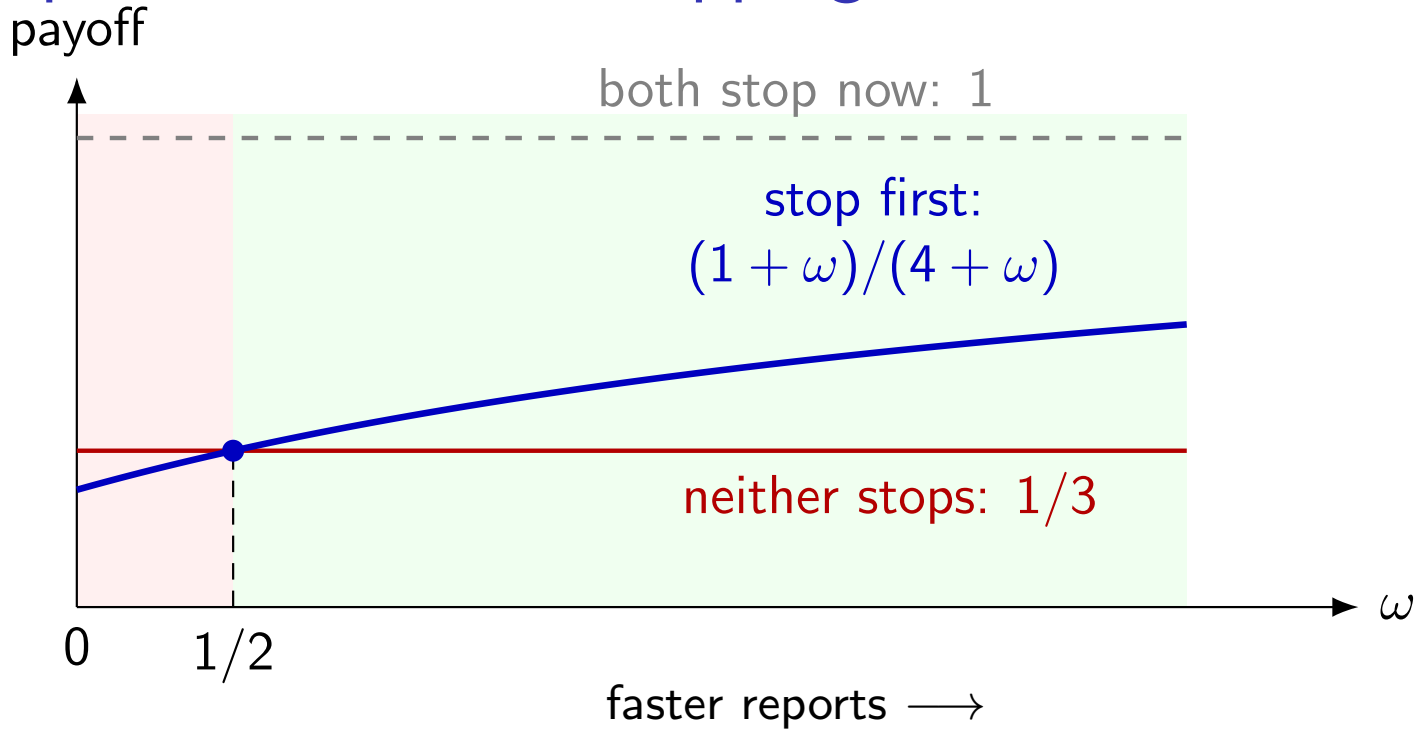


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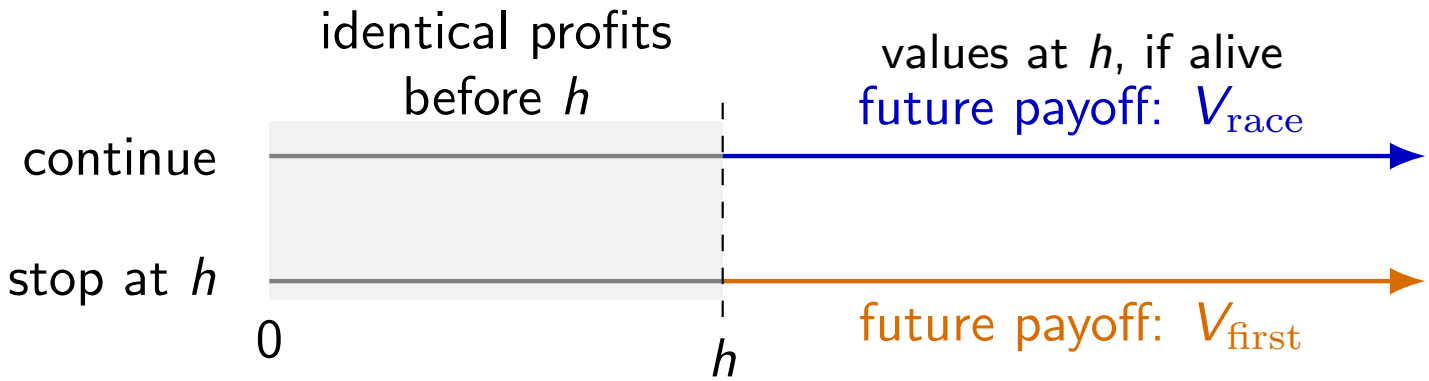
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$$V_{\text{first}} > V_{\text{race}} \iff \frac{1+\omega}{4+\omega} > \frac{1}{3} \iff \omega > \frac{1}{2}.$$

Above 1/2, stopping now breaks the racing equilibrium. At or below 1/2, could stopping *later* or *randomly* help?

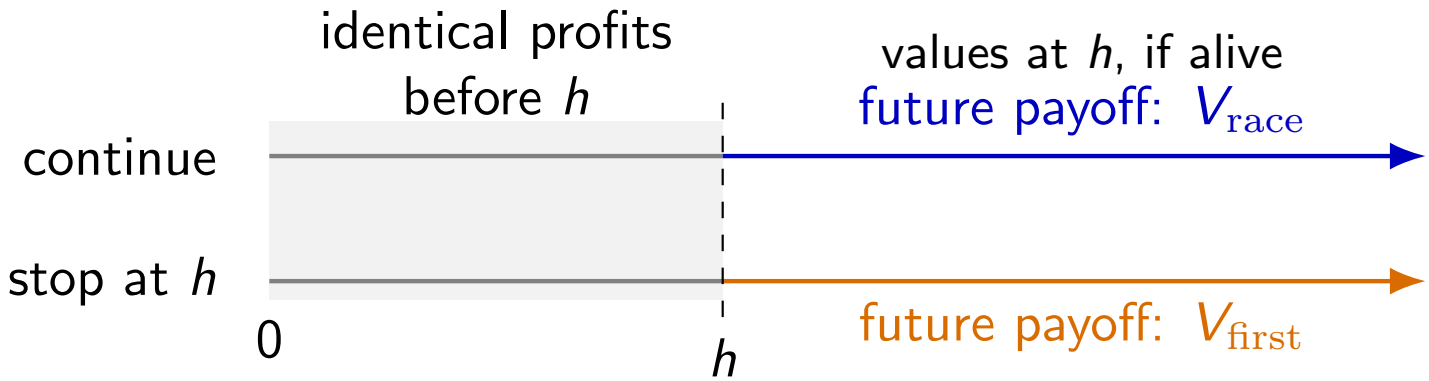
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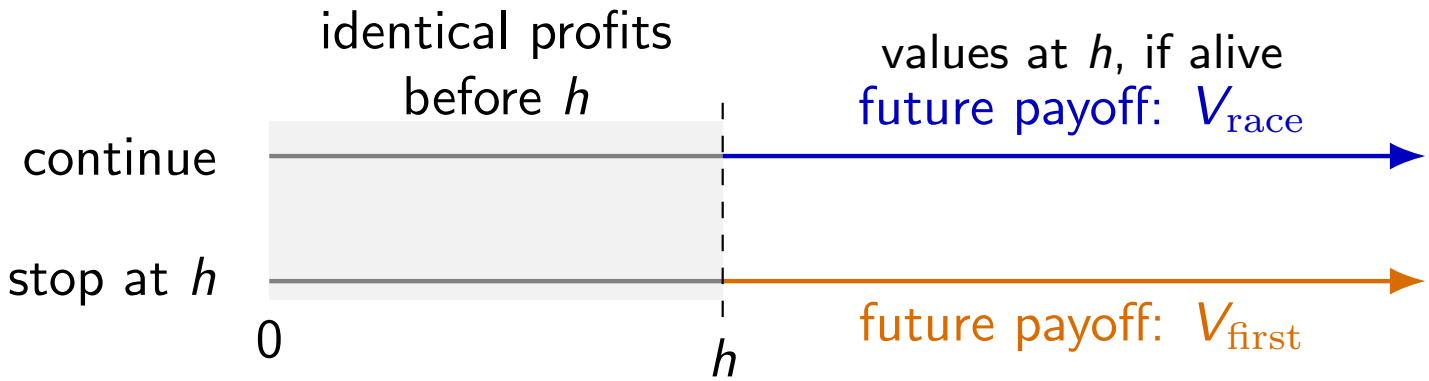
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$$V(h) - V_{\text{race}} = \underbrace{e^{-h}}_{\text{discount}} \quad \underbrace{e^{-2h}}_{\text{survival}} \left(V_{\text{first}} - V_{\text{race}}\right).$$

The multiplier is positive: delaying the stop cannot change whether it helps.

Randomizing the Stopping Time Cannot Help Either

Assume $\omega \leq 1/2$, so $V_{\text{first}} \leq V_{\text{race}} = 1/3$. The preceding comparison gives $V(h) \leq 1/3$ for every finite h; never stopping voluntarily also gives $1/3$.

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The equilibrium conclusion

Stopping upon a report is optimal. Before any report, no stopping plan improves the payoff when $\omega \leq 1/2$; when $\omega > 1/2$, stopping immediately does improve it. This proves the stated equilibrium criterion. □

Safe Stopping and Dangerous Racing Can Coexist

Immediate joint stopping is also a Nash equilibrium, at every monitoring speed. To check a unilateral deviation, hold the rival's action fixed: it stops immediately. Our pointwise comparison then shows that continuing cannot improve our payoff, whether or not we have received its stop report.

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With slow monitoring, cooperation is possible but dangerous racing is stable too. Does fast monitoring rule out perpetual racing in *every* equilibrium?

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For the stronger result, consider Nash equilibria in which each firm stops immediately upon receiving a stop report.

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A *perfect Bayesian equilibrium* requires optimal continuation after every information history, with beliefs updated by Bayes' rule where possible. It therefore satisfies this response requirement.

Fast Monitoring Rules Out Perpetual Racing

Theorem (All equilibria with credible responses)

If $\omega > 1/2$, every Nash equilibrium with stop-on-report responses has finite planned stopping times for both firms, with probability one. In particular, this holds in every perfect Bayesian equilibrium.

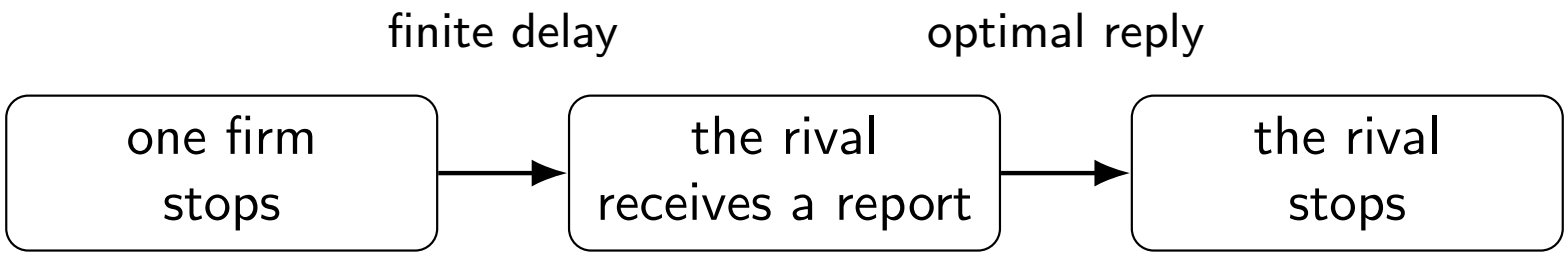
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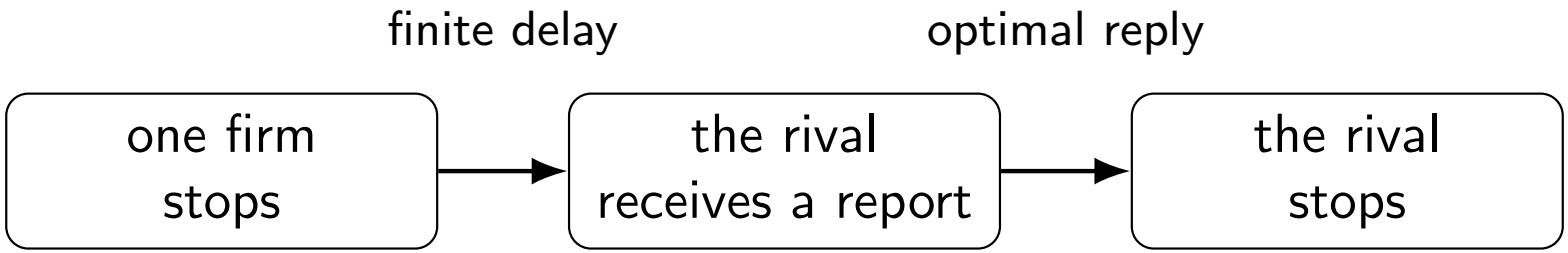


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It is enough to rule out a positive probability that *neither* firm ever stops first. The proof will reuse

$$\Delta := V_{\text{first}} - V_{\text{race}} = \frac{1+\omega}{4+\omega} - \frac{1}{3} > 0.$$

Late but Finite Stopping Plans Become Rare

Fix any equilibrium with credible responses. Draw each firm's private random choices in advance. Let $S_i \in [0, \infty]$ be the date firm i would stop if no report arrived and catastrophe had not ended play. A report can make it stop earlier.

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Suppose, for a contradiction, that

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For each firm, the events $\{t < S_i < \infty\}$ shrink to the empty set: every finite planned date is eventually passed. Therefore

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A feasible deviation at a late date t

Follow the original strategy until t. If still developing and no report has arrived, stop. Otherwise change nothing.

The deviation uses only the firm's own observations, not its rival's unobserved plan S_j.

Stopping Late Has Only One Possible Downside

Compare the deviation with the original strategy when it changes our action at t. Value future payoffs at t, conditional on survival. Recall $\Delta = V_{\text{first}} - V_{\text{race}} > 0$.

Rival already stopped;
its report has not arrived.

→ No loss: stopping is already optimal.

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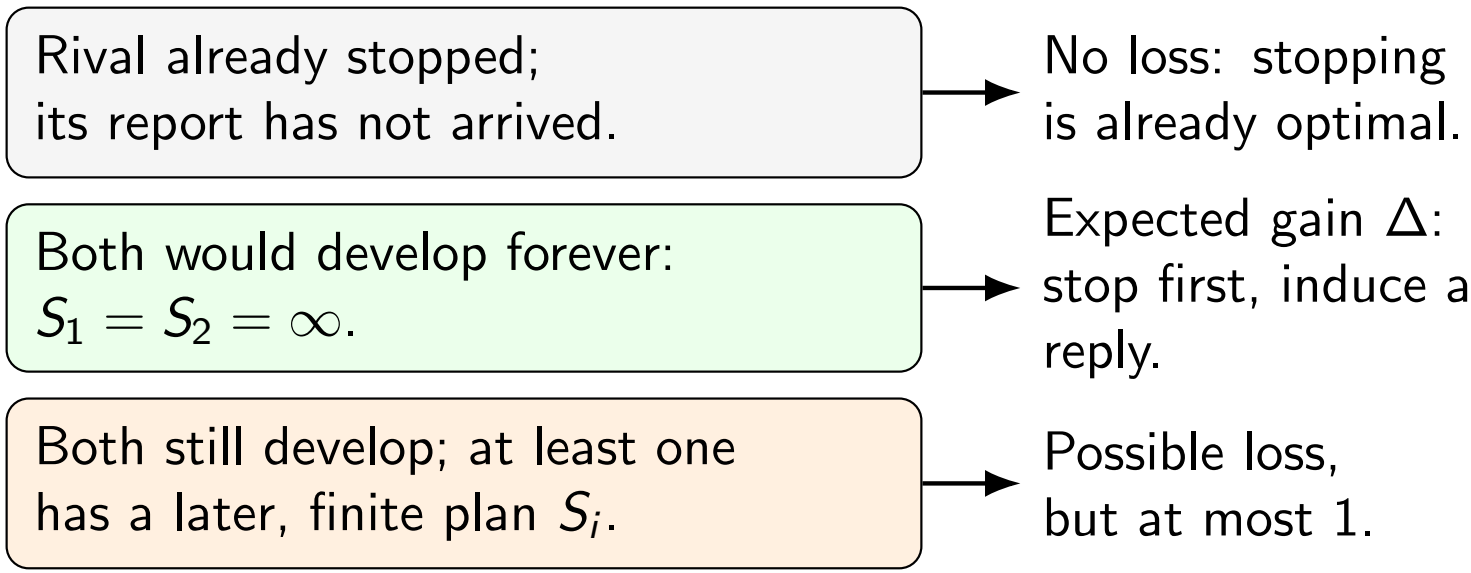
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Both would develop forever: $S_1 = S_2 = \infty$. → Expected gain Δ: stop first, induce a reply.

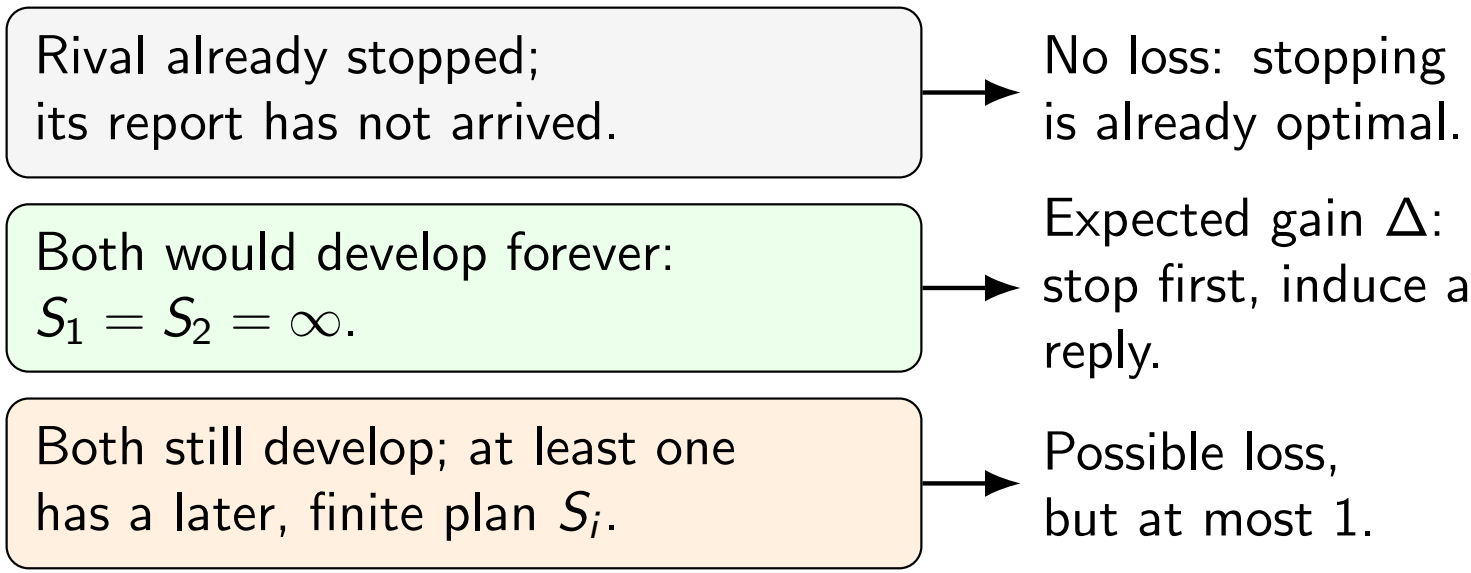
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Why at most 1? In the last case the firms are tied. Freezing the rival at its current capability could only raise our profits and shorten the danger. In that easier problem, our earlier comparison shows that stopping now is optimal and gives 1. Our original continuation is therefore worth at most 1, and the deviation's payoff is nonnegative.

The Gain Eventually Exceeds the Possible Loss

The permanent-racing event has probability $q > 0$ and yields gain $\Delta > 0$. The last case in the diagram has probability at most $\varepsilon_t \to 0$ and costs at most 1.

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Fix the random plans and reporting delays first. On each history where the deviation would change our action at t, our firm develops throughout $[0, t]$. The independent catastrophe process lets us reach t with probability e^{-2t}. Earlier payoffs cancel, so averaging over these histories and discounting back to date zero gives

$$\text{expected payoff gain} \;\geq\; \underbrace{e^{-t}}_{\text{discount}} \quad \underbrace{e^{-2t}}_{\text{survival}} \left(q\Delta - \varepsilon_t\right).$$

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Choose t large enough that $\varepsilon_t < q\Delta$. The deviation is profitable, contradicting Nash equilibrium. Hence $q = 0$: some firm stops in finite time, and its rival stops no later than receiving the report. Both planned stopping times are finite with probability one. $\square$

The Goal Is Safe Progress

Our model allows only continued development or permanent stopping. This isolates how a rival's delayed response affects the incentive to stop first.

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The recurring question

Can competitors observe one another's restraint, and do they have an incentive to respond by reducing their own risk-taking?

Summary

- Firms can all prefer coordinated restraint while sustaining a dangerous race through their individual incentives.

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- Firms can all prefer coordinated restraint while sustaining a dangerous race through their individual incentives.
- Even when the rival will stop after learning of our stop, the expected reporting delay can make stopping first unattractive.
- With credible responses to stop reports, sufficiently fast monitoring rules out perpetual racing in *every* equilibrium.

Thanks!

See you next class!

References

- Drew Fudenberg and Andrew Koh. *Racing to Ruin*. Working paper, August 17, 2026.

  The lecture specializes Example 3 and both directions of Proposition 13 to profit $e^{x_i - x_j}$, discount rate 1, and disaster rate 2.

- Drew Fudenberg and Andrew Koh. *Cooperating Against Catastrophe*. Working paper, September 23, 2026.

  The concluding discussion points to its model of continued development as safety improves.

Backup: Payoff for a Fixed Reporting Delay

We stop while tied; the rival stops after delay $D = d$, if alive. Average over catastrophe risk, holding d fixed. Before the report, expected discounted profits are

$$\int_0^d \underbrace{e^{-s}}_{\text{discount}} \underbrace{e^{-2s}}_{\text{survival}} \underbrace{e^{-s}}_{\text{profit rate}} \, ds = \int_0^d e^{-4s} \, ds.$$

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The two contributions, valued when we first stopped, are therefore

$$\underbrace{\frac{1 - e^{-4d}}{4}}_{\text{before the report}} \quad \text{and} \quad \underbrace{e^{-4d}}_{\text{after the report}}.$$

Backup: Average Over the Reporting Delay

The reporting delay has density $\omega e^{-\omega d}$, so

$$\mathbb{E}[e^{-4D}] = \int_0^\infty e^{-4d} \omega e^{-\omega d} \, dd = \frac{\omega}{4+\omega}.$$

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Averaging the two contributions from the preceding slide gives

$$\text{before:} \quad \mathbb{E}\left[\frac{1-e^{-4D}}{4}\right] = \frac{1}{4}\left(1 - \frac{\omega}{4+\omega}\right) = \frac{1}{4+\omega},$$

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Thus

$$\boxed{V_{\text{first}} = \frac{1}{4+\omega} + \frac{\omega}{4+\omega} = \frac{1+\omega}{4+\omega}.}$$

Return to the payoff comparison.